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Viettel AI Competition

Track 2

Dương Trường Bình – TA

MSc. Dương Đình Thắng



Outline

- **Problem**
- **Data**
- **Evaluation**
- **Baseline**
- **Question**



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Problem

❖ Viettel AI Track 2

• Đang diễn ra 3 Phase - 02/07/2026 - 10/09/2026

Bài 2 – Ontological Reasoning in Medical Knowledge Retrieval

Tổng quan Lịch trình

LỘ TRÌNH

3 Phase

1 PHASE 1 **LIVE**

Vòng 1 - Sơ loại

02/07/2026 → 30/07/2026

Tệp ZIP GPU

2 PHASE 2

Vòng 2 - Sơ khảo

17/08/2026 → 19/08/2026

API endpoint GPU

3 PHASE 3

Vòng 3 - Chung kết

09/09/2026 → 10/09/2026

API endpoint GPU

Problem

❖ From notes to reusable clinical data

FREE-FORM CLINICAL NOTE

1. Tiền sử bệnh

Tăng huyết áp, đái tháo đường type 2.

2. Bệnh sử hiện tại

Không khó thở. Đau ngực khi gắng sức.

3. Thuốc đang dùng

Amlodipine 10 mg po daily.

➔
Information
extraction

STRUCTURED CLINICAL DATA

Symptoms khó thở · đau ngực

Diagnoses tăng huyết áp · đái tháo đường

Medication amlodipine 10 mg

Context negated · historical

DOWNSTREAM USES

- 1 Medical coding
- 2 Patient search
- 3 Clinical timeline
- 4 Analytics & research

❖ Task Definition

Clinical free text

Danh sách thuốc trước nhập viện chính xác và đầy đủ. 1. amlodipine 10 mg po daily 2. aspirin 81 mg po daily 3. metoprolol succinate xl 50 mg po daily 4. guaifenesin ml po q6h:prn điều trị ho 5. nystatin oral suspension 5 ml po qid:prn điều trị đau nhức 6. acetaminophen 325-650 mg po q6h:prn điều trị sốt đau 7. pravastatin 40 mg po daily 8. docusate sodium 100 mg po bid điều trị táo bón 9. senna 8.6 mg po bid:prn điều trị táo bón 10. clonazepam 0.5 mg po qam:prn điều trị lo âu 11. clonazepam 1.5 mg po qhs điều trị lo âu mất ngủ'

Structured medical concepts

```
{  
  "text": "amlodipine 10 mg po daily",  
  "type": "THUỐC",  
  "candidates": ["308135"],  
  "assertions": ["isHistorical"],  
  "position": [58, 83]  
}
```

detect span

01

assign type

02

resolve context

03

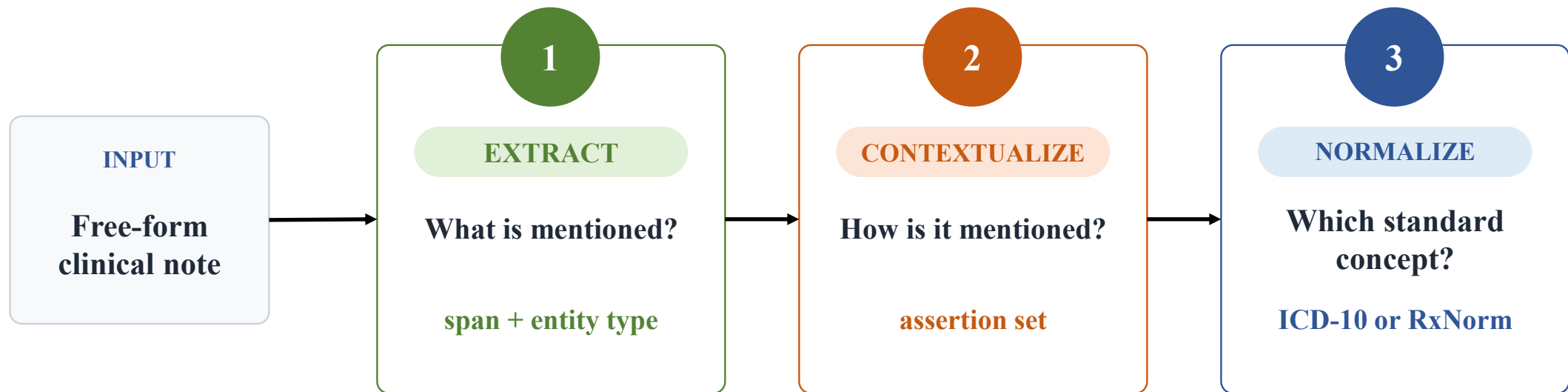
normalize code

04

A valid output object must make all four decisions consistently.

❖ Problem Formulation

Track 2 combines three connected NLP tasks



Each stage adds fields required by the final concept object.

❖ Output Object

```
{  
  "text": "amlodipine 10 mg po daily",  
  "type": "THUỐC",  
  "candidates": ["308135"],  
  "assertions": ["isHistorical"],  
  "position": [58, 83]  
}
```

text

exact substring in the original note

type

one of five concept types

candidates

standard code, when applicable

assertions

context flags, when applicable

position

character offsets

Optional fields are determined by type, not by model convenience.

❖ Concept Types & Required Fields

TYPE	position	assertions	candidates	standard
TRIỆU_CHỨNG	✓	✓	—	—
TÊN_XÉT_NGHIỆM	✓	—	—	—
KẾT_QUẢ_XÉT_NGHIỆM	✓	—	—	—
CHẨN_ĐOÁN	✓	✓	✓	ICD-10
THUỐC	✓	✓	✓	RxNorm RXCUI

position is required for every concept; candidates only apply to diagnosis and drug concepts.

Problem

❖ Character Offsets

```
input[start:end] == text
```

... amlodipine 10 mg po daily ...

start = 58

end = 83

$$83 - 58 = 25$$

The example supports a half-open interval [start, end).

Common failure

normalizing the text and returning offsets from the modified string

❖ Context Assertions

“không khó thở”



isNegated

POLARITY

“tiền sử tăng huyết áp”



isHistorical

TIME

“mẹ bị đái tháo đường”



isFamily

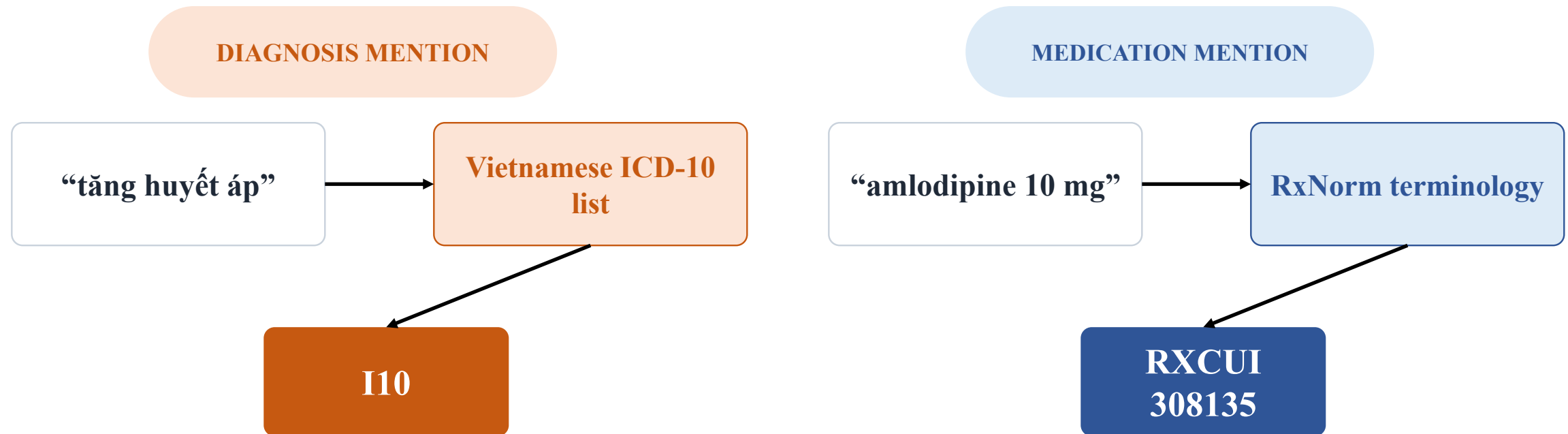
WHO

Assertions change the interpretation of a concept, not its type.

Applicable to TRIỆU_CHỨNG , CHẨN_ĐOÁN , THUỐC

❖ Medical Entity Linking

Entity linking requires 2 external knowledge bases



Competition notes do not include these lookup resources; download and version them separately.



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Data

❖ What Is Provided

100

test notes

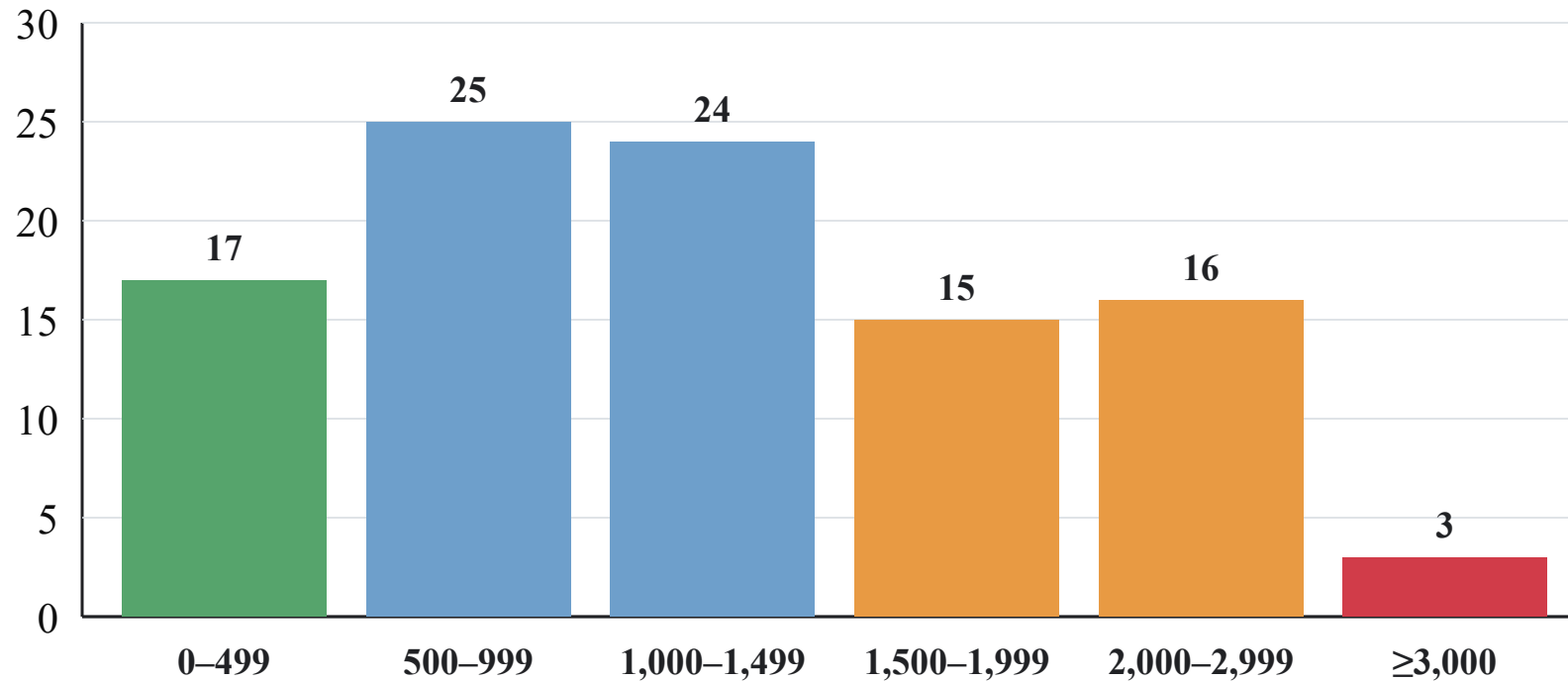
UTF-8 TXT

```
input/  
├── 1.txt  
├── 2.txt  
├── ...  
└── 100.txt
```

Not provided

- ✗ Labeled training data
- ✗ Official local ground truth
- ✗ Vietnamese ICD-10 terminology
- ✗ RxNorm terminology

❖ Length Distribution



number of characters per note

1,223

median characters

136 → 4,428

minimum → maximum

> 30×

length variation

❖ Semi-structured Clinical Notes

1. Tiền sử bệnh

Các tập kinh lâm sàng trước đây

- Tái phát các cơn ngất xỉu trong khoảng vài năm nay.
- Ban đầu các cơn xảy ra vài lần một năm, tăng tần suất...

Các bệnh lý nội khoa mạn tính

- bệnh tim mạch do xơ vữa động mạch
- tăng huyết áp
- phình động mạch chủ nhỏ

2. Bệnh sử hiện tại

Lý do nhập viện: cơn ngất xỉu

Các triệu chứng hiện tại

- Không đánh trống ngực
- Không chóng mặt
- đau ngực · khó thở khi gắng sức

98 / 100

notes with numbered sections

93 / 100

notes with bullet lines

74 / 100

notes with ≥ 3 numbered sections

Structure is a contextual signal, not ground truth.

❖ Context Signals & Text Noise

Surface cues



number of notes containing regex cues

Observed noise

atenololtrong

bình thườngbình thườngbình thường

cảm giáckhó chịu

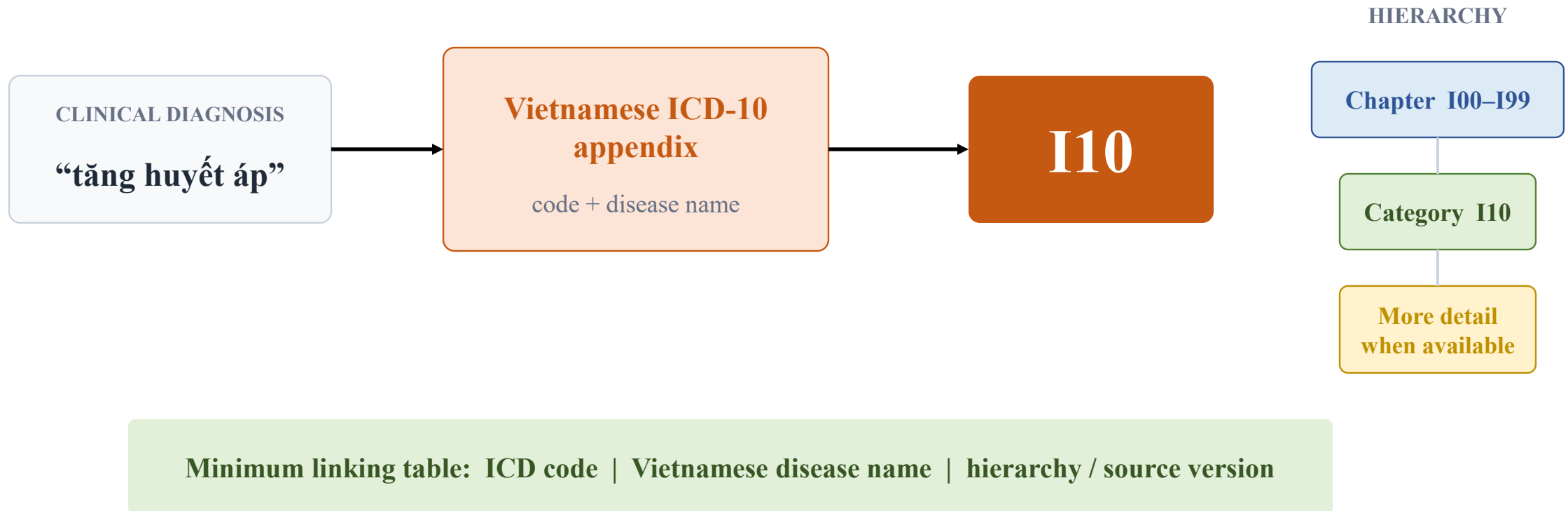
xét nghiệm

EDA REGEX => NOT LABEL DISTRIBUTION

Detect on normalized text if needed, always return offsets on the original note.

❖ External Knowledge Data

ICD-10 is a hierarchical disease classification



❖ External Knowledge Data

<https://dongnaicdc.vn/thong-tu-quy-dinh-ve-ma-hoa-benh-tat-nguyen-nhan-tu-vong-theo-icd-10>

The screenshot shows the website of the Dong Nai City Center for Disease Control. The header features the CDC logo and the text "TRUNG TÂM KIỂM SOÁT BỆNH TẬT THÀNH PHỐ ĐỒNG NAI" and "DONG NAI CITY CENTER FOR DISEASE CONTROL". The navigation bar includes links for Home, Introduction, News - Events, Specialized Activities, Services, Consultation, Video, Information Materials, and Contact. The main content area displays a notice titled "Thông tư quy định về mã hóa bệnh tật, nguyên nhân tử vong theo ICD-10" dated 16/04/2026 09:20. It provides links to download the notice (QĐ BYT.pdf) and two attachments: "Phụ lục 1: Phụ lục Bảng danh mục ICD10_FINAL.xlsx" and "Phụ lục 2: Phụ lục Bảng danh mục ICD10_FINAL.signed.pdf". There are also social media sharing buttons for Facebook, X, Email, and WhatsApp. On the right side, there is a search bar and a "THÔNG BÁO" (Notice) section with two items: "Quyết định Về việc công nhận kết quả tuyển dụng viên chức trúng tuyển đợt 1 năm 2026 của Trung tâm Kiểm soát bệnh tật" and "Thư mời báo giá (lần 2) mua sắm phụ kiện trang phục kiểm dịch y tế năm 2026 của CDC Đồng Nai".

TRUNG TÂM KIỂM SOÁT BỆNH TẬT THÀNH PHỐ ĐỒNG NAI
DONG NAI CITY CENTER FOR DISEASE CONTROL

Thông tư quy định về mã hóa bệnh tật, nguyên nhân tử vong theo ICD-10
16/04/2026 09:20

Tải file Thông tư về tại đây: QĐ BYT.pdf

Phụ lục 1: Phụ lục Bảng danh mục ICD10_FINAL.xlsx

Phụ lục 2: Phụ lục Bảng danh mục ICD10_FINAL.signed.pdf

Share with friends

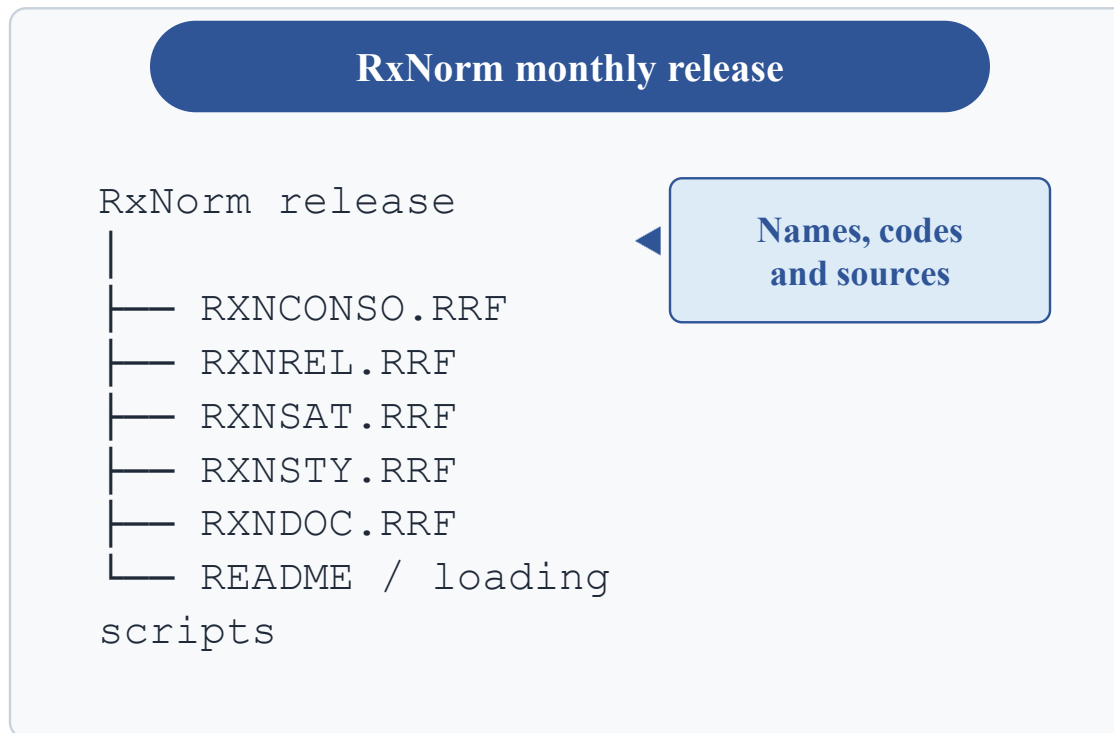
f Chia sẻ X Post Email Chia sẻ

THÔNG BÁO

- > Quyết định Về việc công nhận kết quả tuyển dụng viên chức trúng tuyển đợt 1 năm 2026 của Trung tâm Kiểm soát bệnh tật
- > Thư mời báo giá (lần 2) mua sắm phụ kiện trang phục kiểm dịch y tế năm 2026 của CDC Đồng Nai

❖ External Knowledge Data

The RxNorm download is a package, not a single drug list



For a basic medication linker

**START WITH
RXNCONSO.RRF**

RRF = pipe-delimited UTF-8 text

FULL MONTHLY

broader coverage
license agreement

PRESCRIBABLE

smaller active subset
no license required

❖ External Knowledge Data

<https://www.nlm.nih.gov/research/umls/rxnorm/docs/rxnormfiles.html>

- Create a **UMLS Terminology Services (UTS)** account
- Sign in to the UTS portal and download the latest **RxNorm Full Monthly Release**.
- Extract the ZIP archive and use `rrf/RXNCONSO.RRF` as the RxNorm knowledge-base file.

RxNorm Files

The license agreement required for the RxNorm releases is available from: <https://www.nlm.nih.gov/databases/umls.html>.

The terms and conditions for using RxNorm data is available on the RxNorm Terms of Service page: <https://www.nlm.nih.gov/research/umls/rxnorm/docs/termsofservice.html>.

RxNorm Full Monthly Release July 6, 2026	RxNorm Weekly Updates July 2026
RxNorm_full_07062026.zip MD5 checksum: 33acdc0176af35808f91b3fc74ff2bb4 Release Notes RxNorm Technical Documentation	Current: RxNorm_weekly_07152026.zip Previous: RxNorm_weekly_07082026.zip
Current Prescribable Content Monthly Release July 6, 2026	Current Prescribable Content Weekly Updates July 2026
RxNorm_full_prescribe_07062026.zip <i>(no license required)</i> MD5 checksum: 767678e3b5b1d6fe358b61c21659f3ef Release Notes Current Prescribable Content Documentation	Current: RxNorm_weekly_prescribe_07152026.zip Previous: RxNorm_weekly_prescribe_07082026.zip

❖ External Knowledge Data

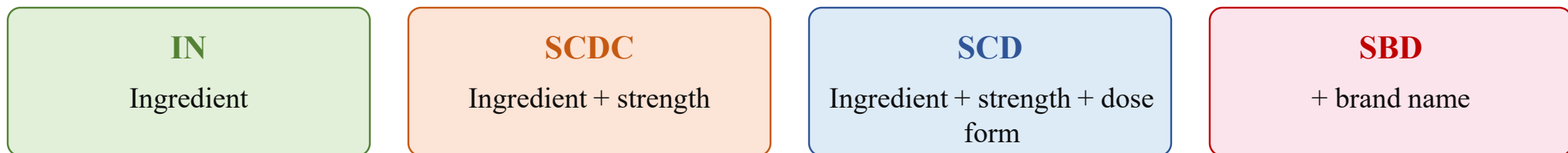
“amlodipine” and “amlodipine 10 mg oral tablet”?

How to read RXNCONSO.RRF

RXCUI	LAT	SAB	TTY	STR	SUPPRESS
310965	ENG	RXNORM	SCD	Ibuprofen 200 MG Oral Tablet	N

- RXCUI**
standard concept identifier
- STR**
drug name represented by the row
- TTY**
concept granularity
- SAB / SUPPRESS**
source and suppressibility

COMMON GRANULARITY LEVELS



❖ Submission Package

INPUT

```
test/  
└─ input/  
    └─ 1.txt      # Văn bản đầu vào của bản ghi 1  
    └─ 2.txt      # Văn bản đầu vào của bản ghi 2  
    └─ ...  
    └─ 100.txt
```

one prediction list per note



OUTPUT

```
output.zip  
└─ output/  
    └─ 1.json  
    └─ 100.json
```

- ✓ 100 files
- ✓ matching names
- ✓ valid JSON
- ✓ original offsets
- ✓ correct folder level



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Evaluation

❖ Final Score

30%

`text_score`

`mean(1 - WER)`

30%

`assertions_score`

`mean(Jaccard)`

40%

`candidates_score`

`weighted Jaccard`

```
final_score = 0.3 × text_score + 0.3 × assertions_score + 0.4 × candidates_score
```

Candidate linking carries the largest official weight.

❖ Text Matching

$$WER = \frac{S + D + I}{N}$$

- *S*: Substitutions
- *D*: Deletions
- *I*: Insertions

Ground truth: amlodipine 10 mg
Prediction: amlodipine 5 mg

$$WER = \frac{1}{3} = 0.333$$

1



2



3

per sample

invert

average

BTC computes $WER(i)$

lower $WER \rightarrow$ higher $1 - WER$

sum and divide by $len(test)$

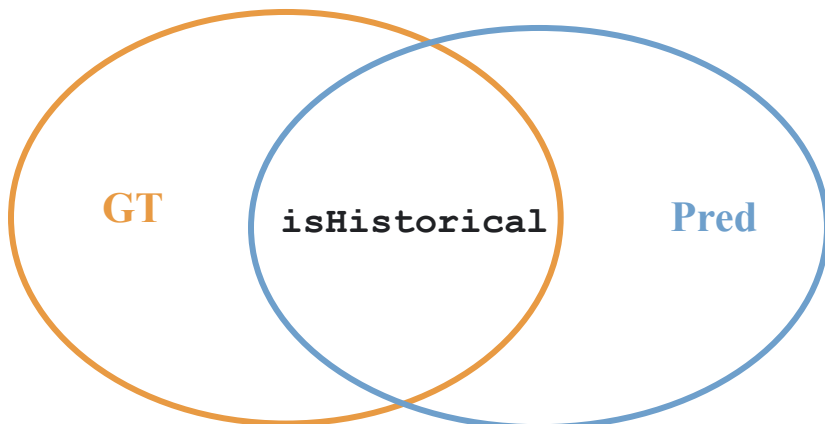
$$\text{text_score} = [\sum_i (1 - WER(i))] / len(test)$$

❖ Assertions Jaccard

1 $GT = \emptyset$ and $Pred = \emptyset$ $J = 1$

2 $GT = \emptyset$ and $Pred \neq \emptyset$ $J = 0$

3 otherwise $J = |GT \cap Pred| / |GT \cup Pred|$



$$\begin{aligned} GT &= \{\text{isHistorical}, \text{isNegated}\} \\ Pred &= \{\text{isHistorical}\} \\ J &= 1 / 2 \end{aligned}$$

❖ Candidates Jaccard

$$\text{candidates_score} = \frac{\sum_i [J_i \times w_i]}{\sum_i w_i}$$

$$w_i = \sum_{k \in i} [\text{len}(\text{ground_truth}(k)) + 1]$$

Sample A

1 GT candidate entry



$$w = 2$$

Sample B

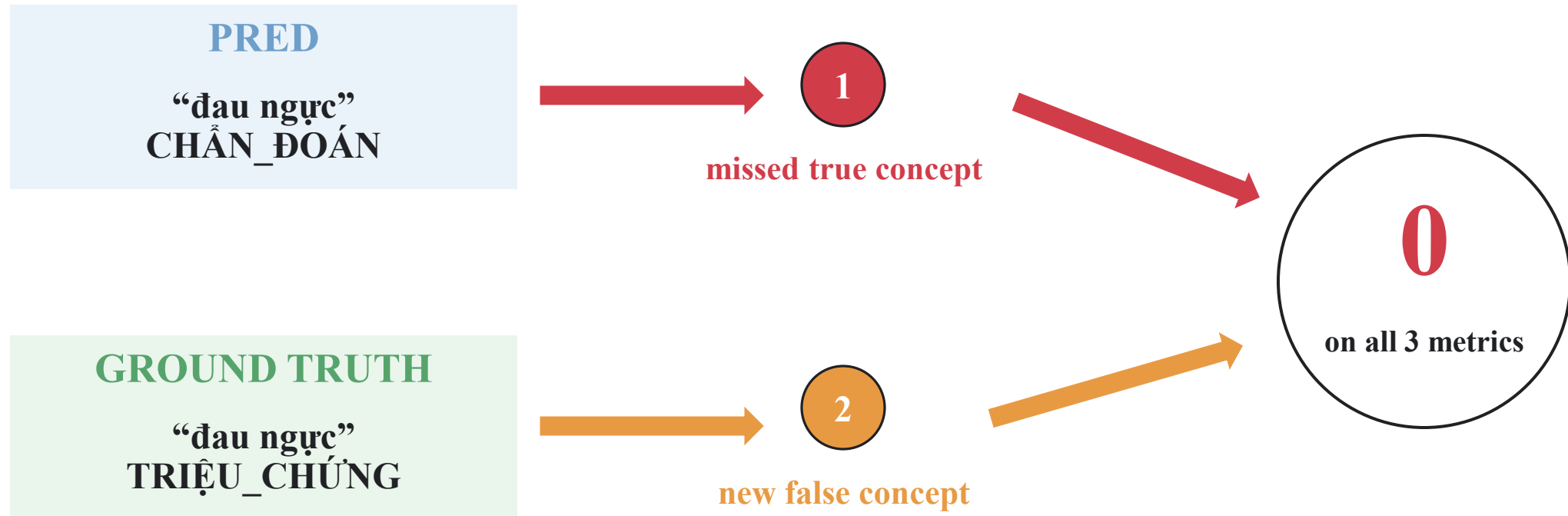
3 GT candidate entries



$$w = 6$$

Direct implication: samples with more GT candidate entries receive more weight.

❖ Type Error Penalty



Correct text + wrong type is counted twice.

❖ Round 1 Constraints

1 PHASE 1 **LIVE**

Vòng 1 - Sơ loại

02/07/2026 → 30/07/2026

Tệp ZIP GPU

2 PHASE 2

Vòng 2 - Sơ khảo

17/08/2026 → 19/08/2026

API endpoint GPU

3 PHASE 3

Vòng 3 - Chung kết

09/09/2026 → 10/09/2026

API endpoint GPU

5×

submissions / day

600s

waiting time

API

self-host only

≤9B

LLM / agent

~15

submit full
package



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Baseline

❖ Task Definition

Clinical free text

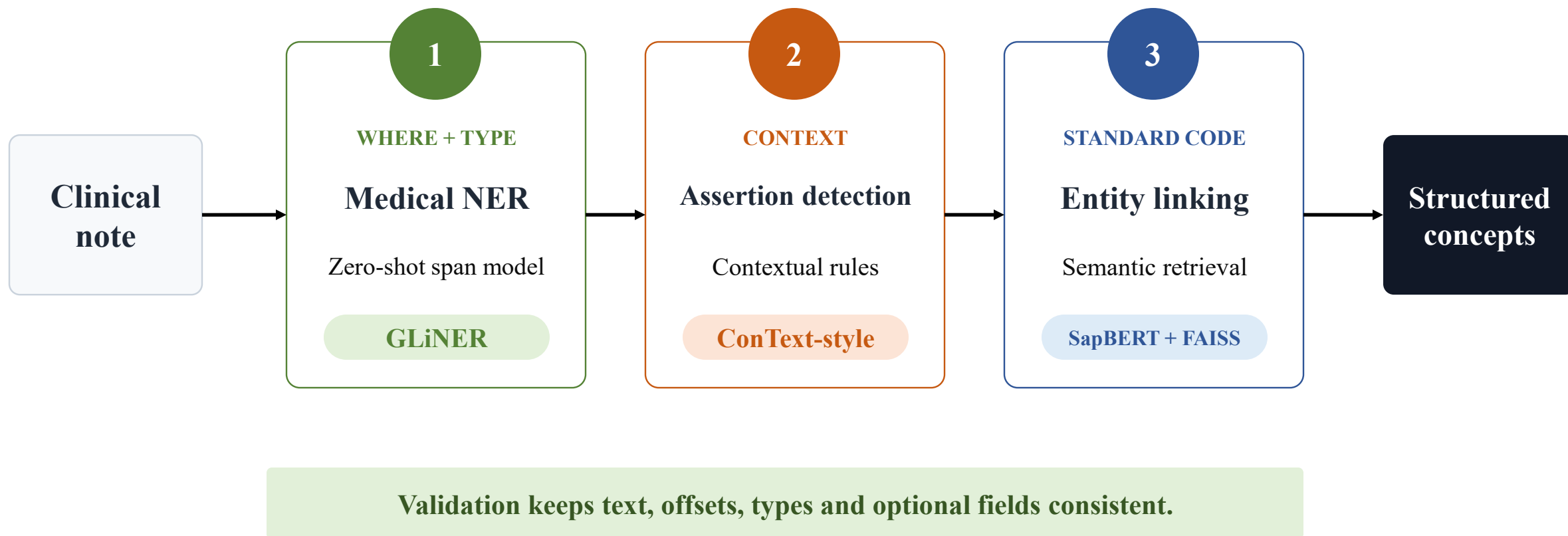
Danh sách thuốc trước nhập viện chính xác và đầy đủ. 1. amlodipine 10 mg po daily 2. aspirin 81 mg po daily 3. metoprolol succinate xl 50 mg po daily 4. guaifenesin ml po q6h:prn điều trị ho 5. nystatin oral suspension 5 ml po qid:prn điều trị đau nhức 6. acetaminophen 325-650 mg po q6h:prn điều trị sốt đau 7. pravastatin 40 mg po daily 8. docusate sodium 100 mg po bid điều trị táo bón 9. senna 8.6 mg po bid:prn điều trị táo bón 10. clonazepam 0.5 mg po qam:prn điều trị lo âu 11. clonazepam 1.5 mg po qhs điều trị lo âu mất ngủ'



Structured medical concepts

```
{  
  "text": "amlodipine 10 mg po daily",  
  "type": "THUỐC",  
  "candidates": ["308135"],  
  "assertions": ["isHistorical"],  
  "position": [58, 83]  
}
```


❖ Overview



❖ Medical NER: The Problem

THE PROBLEM

FREE-FORM CLINICAL NOTE

Bệnh nhân nam 65 tuổi, tiền sử tăng huyết áp và đái tháo đường type 2. Hiện tại không khó thở. Thuốc đang dùng: amlodipine 10 mg.

Each note is unstructured prose. Diagnoses, symptoms and drugs are scattered through the text — we cannot standardize or reason about them until we know exactly where each concept sits.

WHAT WE NEED

- 1 Locate the span — the exact character range of every medical concept.
- 2 Assign the type — CHẨN_ĐOÁN, TRIỆU_CHỨNG, THUỐC, XÉT_NGHIỆM ...

What is GLiNER?

A zero-shot span model. Given only the label names as a prompt, it finds and tags entities with no labeled training data - essential here, since the competition provides no training set.

❖ Span Detection

INPUT: clinical text

Bệnh nhân không **khó thở**, đang dùng **amlodipine 10 mg**.

OUTPUT: spans + types

[16, 23) “khó thở”

TRIỆU_CHỨNG

[35, 51) “amlodipine 10 mg”

THUỐC

NER answers two questions

1

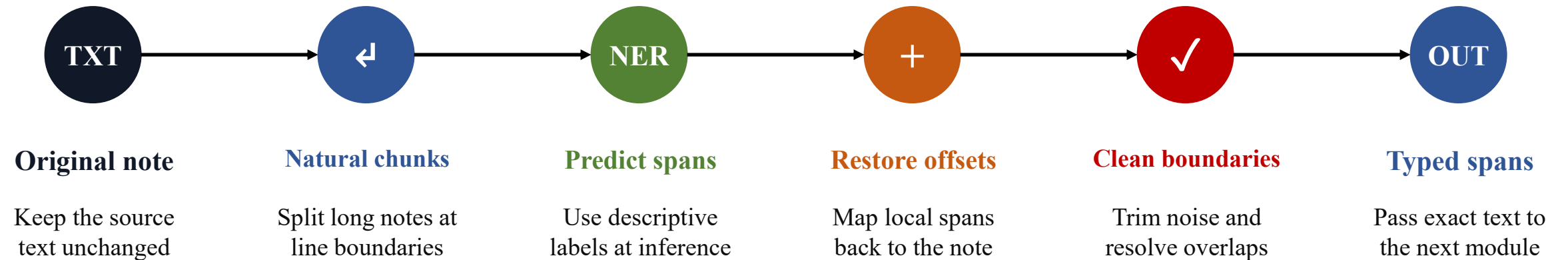
Where does the concept begin and end?

2

Which of the five entity types is it?

No labeled train set → use zero-shot NER

❖ Span Detection



Invariant: `input[start:end] == text`

Method used: GLiNER zero-shot span NER

❖ Assertion Detection: The Problem

THE PROBLEM

<i>“không khó thở”</i>	→	NEGATED
<i>“tiền sử tăng huyết áp”</i>	→	HISTORICAL
<i>“mẹ bị đái tháo đường”</i>	→	FAMILY

In every row the span is correct, yet the clinical fact is the opposite of the surface words. Read literally, we would record active dyspnea, a current hypertension diagnosis, and the patient's own diabetes, none of which are true.

WHAT WE NEED

- 1 Detect polarity: is the concept affirmed or negated?
- 2 Detect temporality & experiencer: current vs. past, patient vs. family.

What is ConText?

A lightweight rule-based algorithm. For each concept it scans a few words around the span for trigger cues - e.g. “không”, “tiền sử”, “mẹ” - and flips the assertion accordingly. Fast, transparent, and needs no training data.

❖ Context Understanding



“không khó thở nhưng đau ngực” ?

❖ Entity Linking

WHY EXACT MATCH FAILS

amlodipine 10 mg

amlodipin 10mg

Norvasc 10 mg

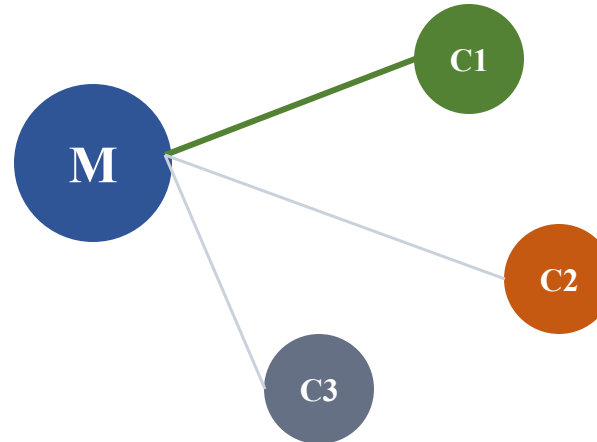
AML 10

Synonyms, spelling variants, trade names and dose forms may refer to the same concept.

What are embeddings?

Text turned into vectors of numbers, so phrases with similar meaning sit close together.

SEMANTIC SIMILARITY



Nearby vectors represent similar biomedical meaning

What is FAISS?

A library that searches millions of vectors in milliseconds to return the nearest matches.

STANDARDIZED OUTPUT

tăng huyết áp

ICD-10 → I10

amlodipine 10 mg

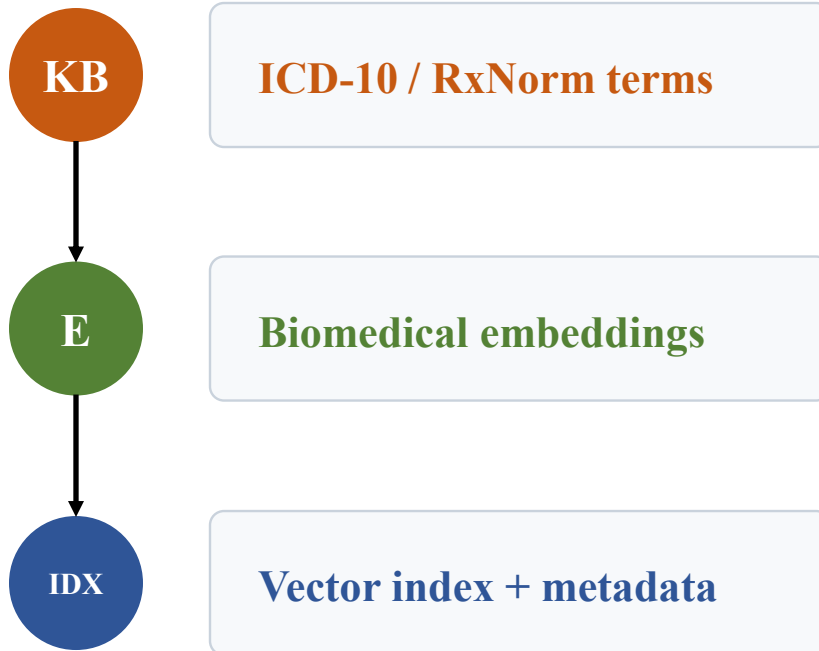
RxNorm → 308135

What is SapBERT?

A biomedical language model whose embeddings are tuned to recognise medical synonyms.

❖ Candidate Retrieval

OFFLINE: prepare the knowledge base once



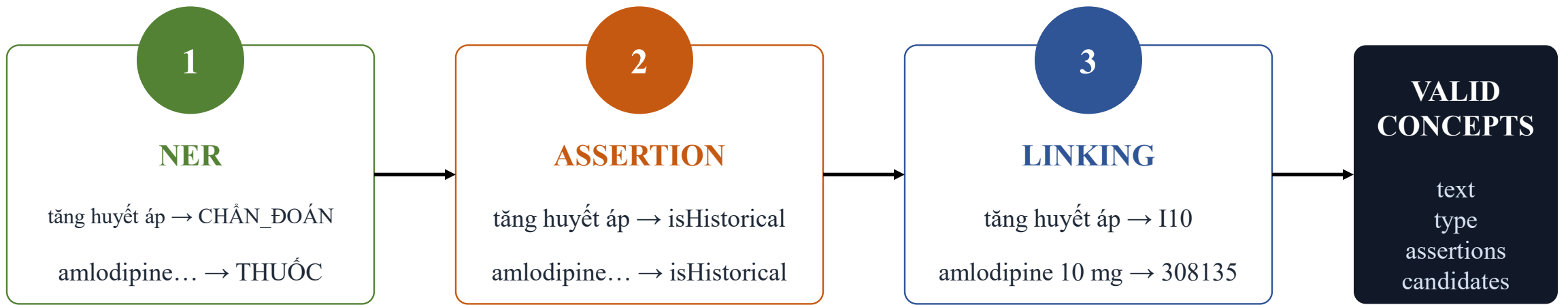
ONLINE: process each detected mention



Input: mention + type → Output: ranked ICD-10 or RxNorm candidates

❖ Example

Tiền sử **tăng huyết áp**. Thuốc trước nhập viện: **amlodipine 10 mg po daily**.



Final validation checks offsets, allowed fields and output order.

❖ Assessment

Strengths

- ✓ No competition labels required
- ✓ Modules can be replaced independently
- ✓ Context rules are transparent
- ✓ Semantic retrieval handles variants

Predictable limitations

- ! Zero-shot NER may miss or over-detect spans
- ! Boundary and type errors reach every later step
- ! Implicit or long-range context is difficult
- ! Linking depends on knowledge-base coverage

NER error

Wrong context

Wrong / no code





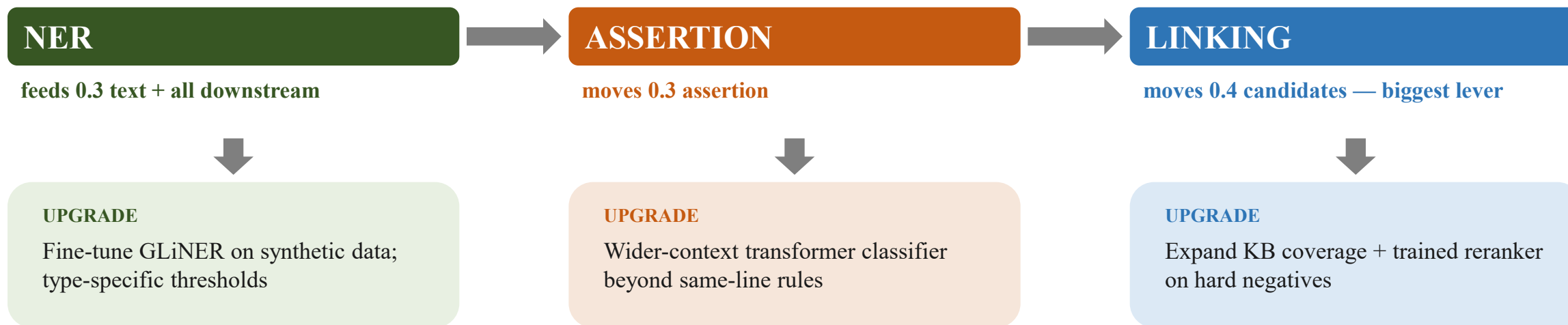
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Next Steps

❖ Roadmap

The baseline is zero-shot everywhere. The next gains could come from (a) creating the labeled data we currently lack, and (b) spending it where the score weight is highest, so we might order by leverage rather than pipeline position.

$$\text{Score} = 0.3 \cdot \text{text} + 0.3 \cdot \text{assertion} + 0.4 \cdot \text{candidates}$$



CROSS-CUTTING ENABLERS

- ① Data synthesis — builds the labeled Vietnamese notes that unlock every fine-tuning step above.
- ② Trustworthy eval + one-change A/B loop — a host-aligned local scorer so each upgrade is measured, not guessed.

❖ NER Enhancements

MOVES: 0.3 · text + downstream

NER precision feeds every stage after it, which is the highest-reach place to improve.

1 Fine-tune GLiNER on synthetic data

Turn the zero-shot model into a supervised one using the notes from the data-synthesis step, the single biggest precision & recall lever.

2 Type-specific thresholds

Calibrate the confidence cut-off per concept type: symptoms over-fire, tests under-fire. One global threshold leaves points on the table.

3 Smarter chunking + span-boundary refiner

Token-budget windows instead of character splits, plus a refiner that snaps predicted spans to exact character boundaries.

4 Self-consistency / light ensemble

Combine multiple passes behind a precision guard, since the metric punishes over-emission hardest, so favour agreement over raw recall.

❖ Data Synthesis: Four Strategies

WHY WE NEED TRAINING DATA

Zero-shot GLiNER + rules are near their ceiling. To fine-tune anything we need labeled Vietnamese clinical notes, which barely exist publicly and are privacy-gated. So we generate them, with labels attached by construction.

#	Strategy	How it works	Where the labels come from
1	Label-first / ontology-driven (most reliable)	Sample real ICD-10 + RxNorm concepts, then prompt an LLM to write a realistic Vietnamese note around them	Inserted by construction → gold type + code + position are free
2	Cross-lingual transfer	Machine-translate annotated English clinical NER corpora (i2b2 / n2c2-style) to Vietnamese; project span labels via alignment	Reuse existing English gold labels
3	Distant / weak supervision	Dictionary-match ontology terms in unlabeled Vietnamese medical text (guidelines, forums) to auto-label	Ontology string match (noisy)
4	Assertion-targeted	Template negation / family / history variants — “không ...”, “mẹ bị ...”, “tiền sử ...”	Template slot = gold assertion

❖ Assertion Enhancements

MOVES: 0.3 · assertion

A correct span with the wrong context is a wrong clinical fact — these upgrades widen the model's view beyond the current line.

1

Transformer assertion classifier

Replace or augment the same-line rules with a small classifier that reads a wider window — directly attacks the implicit / long-range context limitation.

2

Section-structure parsing

Use headings and blank-line blocks to set scope: a “Tiền sử gia đình” (family history) section flips the experiencer for every concept beneath it.

3

LLM assertion with careful prompting

A powerful option — but it must beat the transparent, zero-cost rules on real data to justify the added latency and unpredictability.

❖ Linking Enhancements: The 0.4 Lever

MOVES: 0.4 · candidates: the biggest lever

Linking is 40% of the score and the baseline's weakest module - the most points live here.

1 Expand KB coverage + synonym enrichment

Directly fixes the “no code” failure (limitation #4): more concepts and surface forms in the knowledge base means fewer un-linkable mentions.

2 Trained cross-encoder reranker

A learned upgrade to the LLM reranker, trained on hard negatives mined from the retrieval stage to sharpen the final pick.

3 LLM rerank refinements

Self-consistency voting, a wider retrieve_k, and richer context in the prompt to raise the ceiling of the current reranker.

4 Learn the organizer's conventions

Systematically match the host's linking rules via one-change-at-a-time A/B, since the leaderboard is the only non-circular signal.



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Question

